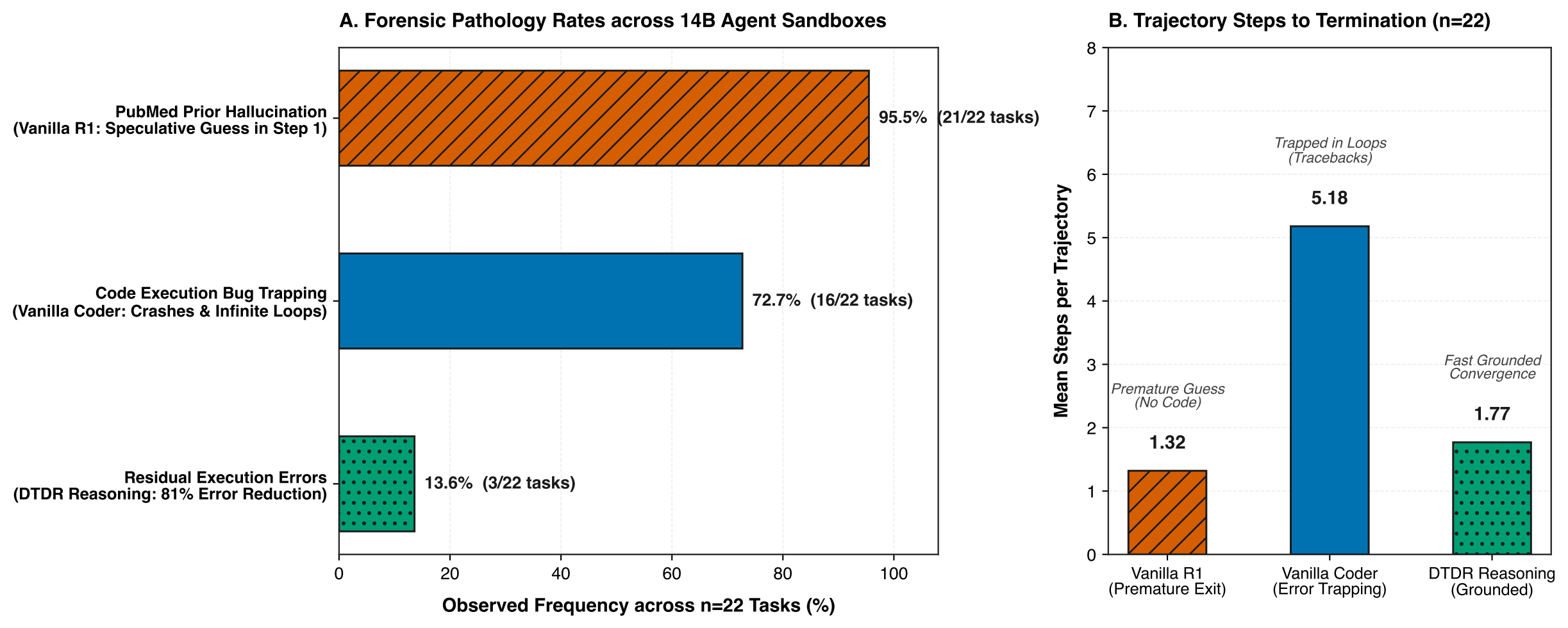

A. Forensic Pathology Rates across 14B Agent Sandboxes
PubMed Prior Hallucination
(Vanilla R1: Speculative Guess in Step 1)
95.5% (21/22 tasks)
Code Execution Bug Trapping
(Vanilla Coder: Crashes & Infinite Loops)
72.7% (16/22 tasks)
Residual Execution Errors
(DTDR Reasoning: 81% Error Reduction)
13.6% (3/22 tasks)
0
20
40
60
80
100
Observed Frequency across n=22 Tasks (%)
B. Trajectory Steps to Termination (n=22)
Mean Steps per Trajectory
0
1
2
3
4
5
6
7
8
Premature Guess
(No Code)
1.32
Trapped in Loops
(Tracebacks)
5.18
Fast Grounded
Convergence
1.77
Vanilla R1
(Premature Exit)
Vanilla Coder
(Error Trapping)
DTDR Reasoning
(Grounded)